

Chapter 5: Multiplexing and Switching

Multiplexing

In general, a medium can carry only one signal at any moment in time. For multiple signals to share one medium, the medium must somehow be divided, giving each signal a portion of the total bandwidth. Multiplexing (also known as MUXing) is a method by which multiple analog message signals or digital data streams are combined into one signal over a shared medium. It works on physical layer of OSI model.



Applications of multiplexing

- Telephone System
- Telemetry
- Broadcasting (Radio and TV)
- Satellites

Advantages of Multiplexing

- More than one signal can be sent over a single medium or link
- Effective use of the bandwidth of medium

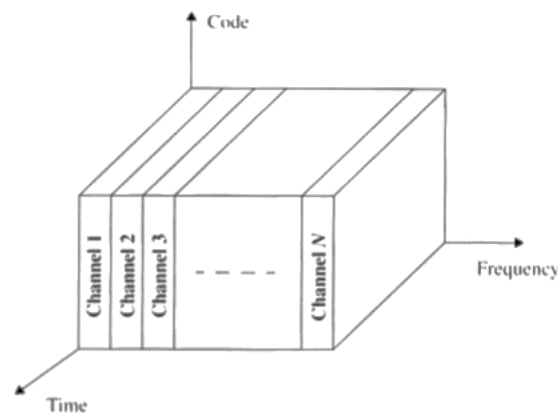
Multiple Access Technique

Application of multiplexing is to help users to get data over the same channel. These techniques ensure that data packets are sent and received efficiently, without collisions or interference. They help manage the network traffic so that all devices can share the communication channel smoothly and effectively. It works on the data link layer of OSI model. If there is a dedicated link between the sender and the receiver then data link control layer is sufficient, however if there is no dedicated link present then multiple stations can access the channel simultaneously.

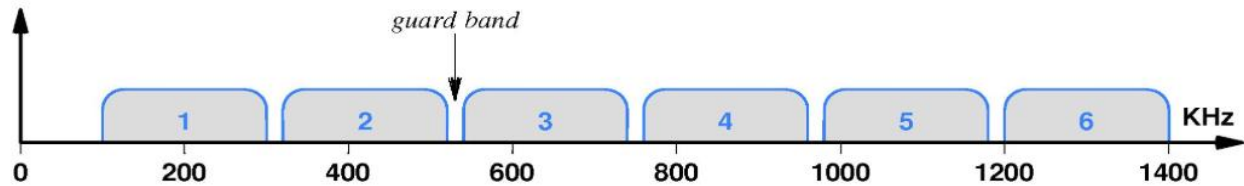
Frequency Division Multiple Access (FDMA)

Each user is assigned a unique frequency band for the duration of the message. In **Frequency-Division Multiple Access (FDMA)**, the total available channel bandwidth is divided into multiple distinct sub-channels. Each user is assigned a specific sub-channel, allowing multiple users to transmit voice and data simultaneously. This method enables simultaneous communication by efficiently sharing the overall bandwidth among several users.

The FDMA technique can be applied to both digital and analog systems, although FDMA is widely used in analog communication systems.



- The channels are assigned on demand by the users
- The same frequency is not allotted for two different users
- The bandwidth of FDMA channel is relatively narrow



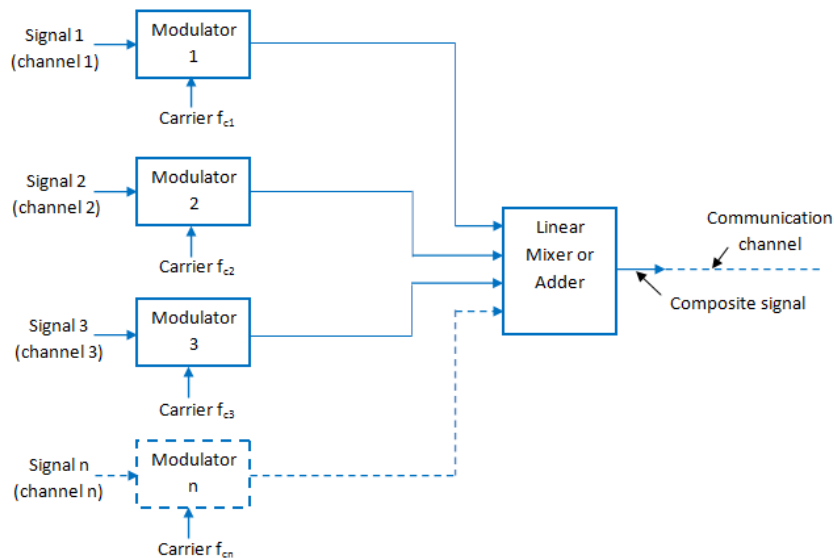
A narrow unused frequency band is left between the different used frequency channels. This is known as guard band. This is used to avoid interference from the adjacent channels.

Number of channels in FDMA

Let B_{total} be the total system bandwidth, B_{guard} be the guard band at edge, and B_{ch} the single radio channel bandwidth. Then the number of channels in FDMA system:

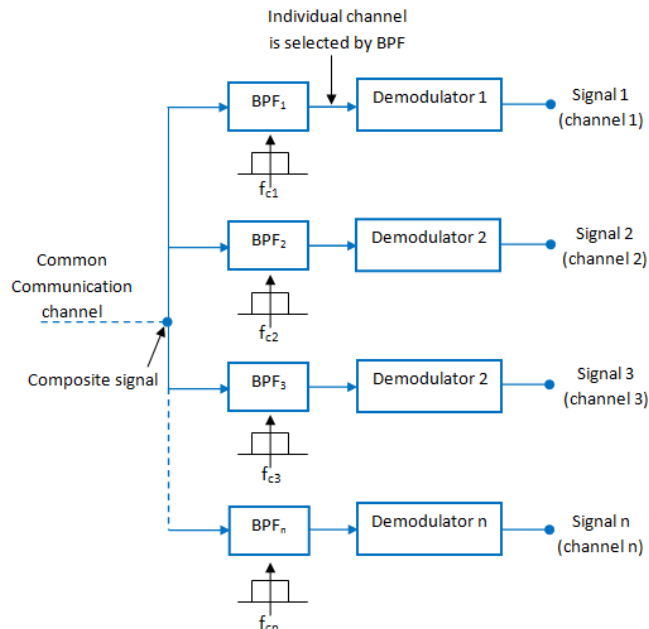
$$N = \frac{B_{total} - 2B_{guard}}{B_{ch}}$$

FDMA Transmitter



The signals that are to be multiplexed will each modulate a separate carrier. The type of modulation can be AM, SSB, FM, or PM. The modulated signals are then added together to form a complex signal, which is transmitted over a single channel.

FDMA Receiver



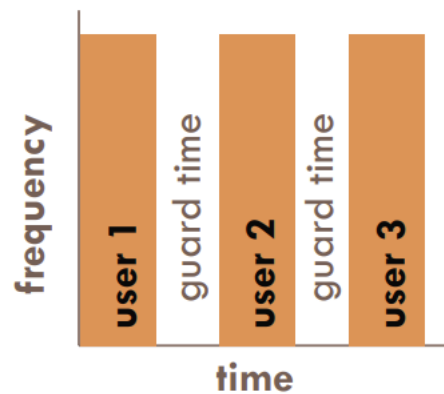
The composite signal is applied to a group of band-pass filters (BPF). Each BPF has a center frequency corresponding to one of the carriers. The BPFs have an adequate bandwidth to pass all the channel information without any distortion. Each filter will pass only its channel and reject all the other channels. The channel demodulator then removes the carrier and recovers the original signal.

FDM Hierarchy

Number of Voice Channels	Bandwidth	Spectrum	AT&T	ITU-T
12	48 kHz	60–108 kHz	Group	Group
60	240 kHz	312–552 kHz	Supergroup	Supergroup
300	1.232 MHz	812–2044 kHz		Mastergroup
600	2.52 MHz	564–3084 kHz	Mastergroup	
900	3.872 MHz	8.516–12.388 MHz		Supermaster group
$N \times 600$			Mastergroup multiplex	
3,600	16.984 MHz	0.564–17.548 MHz	Jumbogroup	
10,800	57.442 MHz	3.124–60.566 MHz	Jumbogroup multiplex	

Time Division Multiple Access (TDMA)

Time Division Multiple Access (TDMA) is a digital modulation technique used in digital cellular telephone and mobile radio communication. In simplest terms, TDMA enables multiple users to share the same frequency by dividing each cellular channel into different time slots. In effect, a single frequency supports multiple and simultaneous data channels. So, with a two-time slot TDMA, two users can share the same frequency. With a three-time slot TDMA, three users can share the same frequency and so on.



- In TDMA, users share same frequency band in non-overlapping time intervals. Eg. Round Robin
- Receiver filters are just windows instead of band-pass filter as in FDMA
- Guard time can be as small as the synchrononization of the network permits. All users must be synchronized with base station to within a fraction of guard band. Guard time of 30-50 μ s common in TDMA.
- Used in GSM, NA-TDMA, Pacific Digital Cellular (PDC)

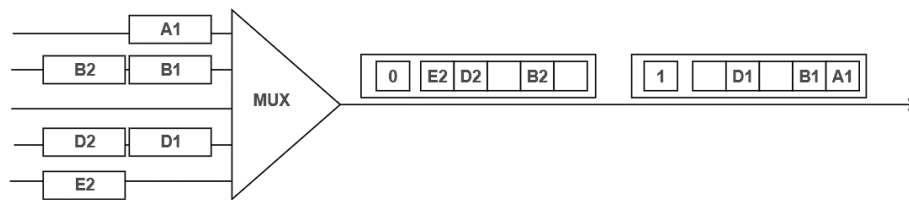
One of the disadvantage of TDMA is that each user has a predefined time slot. However, users roaming from one cell to another are not allotted a time slot. Thus, if all the time slots in the next cell are already occupied, a call might well be disconnected.

1. Synchronous Time Division Multiplexing

Each user is assigned a **fixed, pre-determined time slot**, whether or not they have data to send.

Key Characteristics:

- Even if a user has no data, their time slot remains unused (wasted bandwidth).
- Simple and easy to implement.
- Suitable for constant data streams, like voice.



Advantages of Synchronous TDM:

- Synchronous TDM ensures steady and continuous data transmission by providing each data stream with a fixed time slot, guaranteeing bandwidth.
- Fixed time slots ensure lower delays and, therefore, predictable data delivery times.
- Time slots are pre-allotted; therefore, synchronization between multiple streams of data is easier to manage.
- These are perfect for those applications that require a steady stream of data for smooth and continuous transmission.

Disadvantages of Synchronous TDM

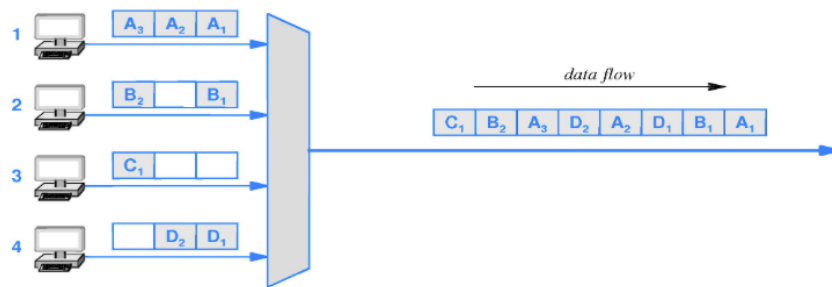
- Time slots are fixed in number, and if any data stream has nothing to send, its slot remains unused. Hence, bandwidth is wasted.
- Synchronous TDM works poorly when there is irregular transmission, or in bursts, because of no scope for dynamic allocation of slots due to demand.
- Higher cost even in unutilized slots

2. Statistical Time Division Multiplexing

Statistical TDM is a type of multiplexing where time slots are dynamically allocated to users only when they have data to send, rather than assigning fixed slots.

Key Characteristics:

- Dynamic slot allocation
- Efficient Bandwidth use
- Buffering required: Since data arrival is unpredictable, buffers and queues are used to store data temporarily.



This method is used in applications where the data rate is variable and unpredictable. Statistical TDM is helpful when input devices do not have a constant data flow.

What is T1?

T1 is a North American digital communication operator standard, and is a digital carrier signal. Its data rate is approximately 1.544 Mb/s. The T1 line operates using two pairs of wires; one for transmission and one for reception, and employs Time Division Multiplexing (TDM) to combine 24 channels, each carrying 64 Kbps of voice or data. Each standard T1 frame consists of 193 bits: 24 voice samples of 8 bits each plus 1 synchronization bit, with 8,000 such frames sent every second.

The basic T1 carrier features are:

- Transmission of signals at the rate of 1.544 Mbps, here is how it is calculated:

◆ T1 carries 24 channel

- ◆ 24 channel for voice/data + It has 1 framing bit
- ◆ 1 Channel = 8bits
- ◆ Total bit=(8 *24)+1=192+1=193
- ◆ Sampling frequency = 8KHz
- ◆ Data rate per channel = 8*8000 =64Kbps
- ◆ (64*24=1536)+(1*8=8)=1536+8=1544=1.544Mbps
- T1 media types including twisted pair cable, coaxial cable, microwave radio, fiber optic cable, and satellite.
- A T1 line is point-to-point. T1 lines may be used fractionally or at their full bandwidth.
- T1 frame consists of 24 timeslots.

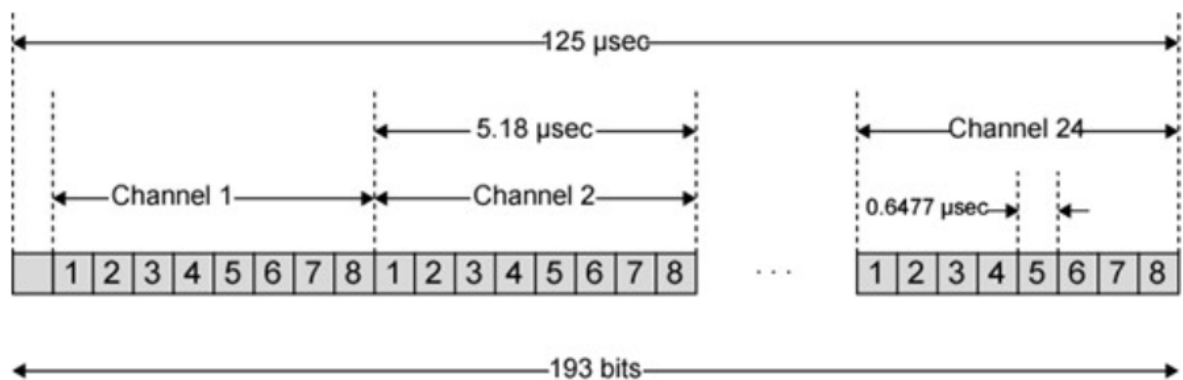


Fig: T1 Frame Format

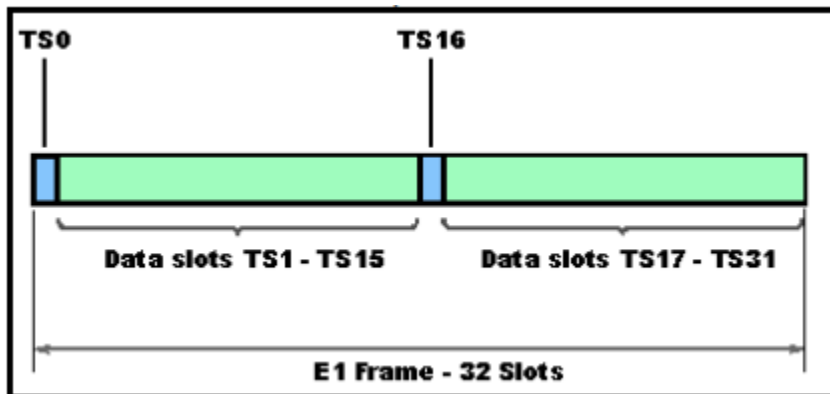
What is E1?

The E1 line uses two wire pairs (one for sending, one for receiving) and time division multiplexing (TDM) to transmit 32 channels 64-Kbps voice or data. It is a 2.048 Mbps point-to-point dedicated, digital circuit provided by the telephone companies in Europe.

The basic E1 Carrier features are shown as below:

- E1 is a digital communication link that enables the transmission of voice, data, and video signals at the rate of 2.048 million bits per second (Mbps). Here is how it is calculated:

- ◆ E1 carries 30+2 channel.
- ◆ 30 channel for voice/data, 2 channel for controlling,
- ◆ 1channel=8 bits
- ◆ Total bit=8*32=256
- ◆ Sampling frequency=8kHz
- ◆ Data rate per channel= 8*8000=64Kbps
- ◆ 32*64=2048=2.048Mbps
- Deployed primarily in Europe and Asia.
- E1 frame consists of 32 timeslots.
- E1 specifications defined in CCITT Recommendation G.704, although Recommendation G.732 supplements G.704.
- Data is sent over one signal pair and simultaneously received on another signal pair. (full duplex transmission).



Digital Subscriber Line (DSL)

The Digital Subscriber Line (DSL), originally known as the digital subscriber loop, is a communication technology that delivers internet services over standard copper telephone lines. Alongside cable internet, DSL is one of the most widely used methods for providing broadband access by internet service providers (ISPs).

Types of DSL

- Symmetric DSL (SDSL)

SDSL splits the upstream and downstream frequencies evenly, providing equal speeds to both uploading and downloading data transfer. This connection may provide 2 Mbps upstream and downstream. It is mostly preferred by small organizations.

- Asymmetric DSL (ADSL)

ADSL provides a wider frequency range for downstream transfers, which offers several times faster downstream speeds. An ADSL connection may offer 20 Mbps downstream and 1.5 Mbps upstream, it is because most users download more data than they upload.

Asymmetric DSL (ADSL)

ADSL is a high-speed internet access service that utilizes existing copper telephone lines to send and receive data at speeds that far exceed conventional dial-up modems.

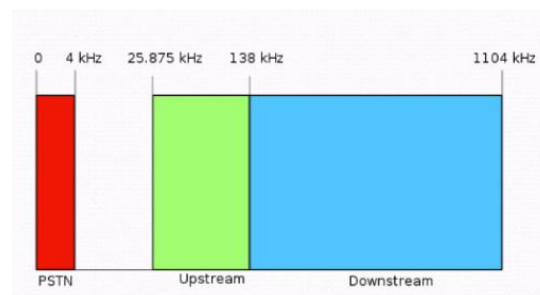
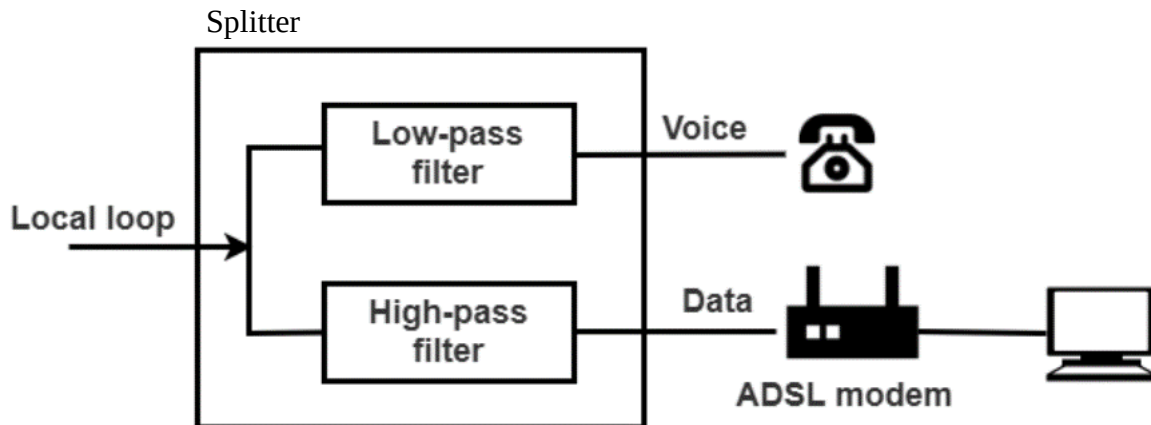


Fig: Frequency plan for ADSL

- Red area is the frequency range used by normal voice telephony, the green (upstream) and the blue (downstream) areas are used for ADSL.

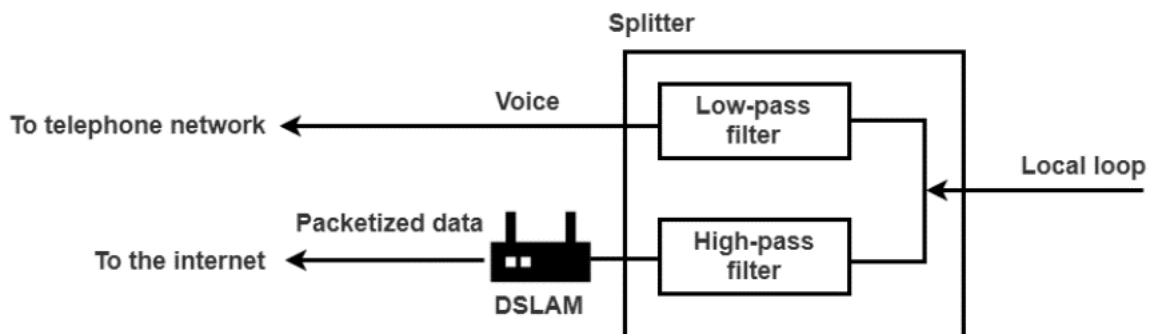
Unfortunately, 1.1 MHz is just a theoretical bandwidth. The factors like distance between residence and switching offices will affect the bandwidth due to reasons like the size of the cable, signal used and so on.

Customer Site: ADSL Modem



The ADSL Modem is installed at a customer's site. In this local loop, it connects to a splitter, which is used to separate voice and data communication. Modulation and Demodulation are performed, creating separate downstream and upstream channels.

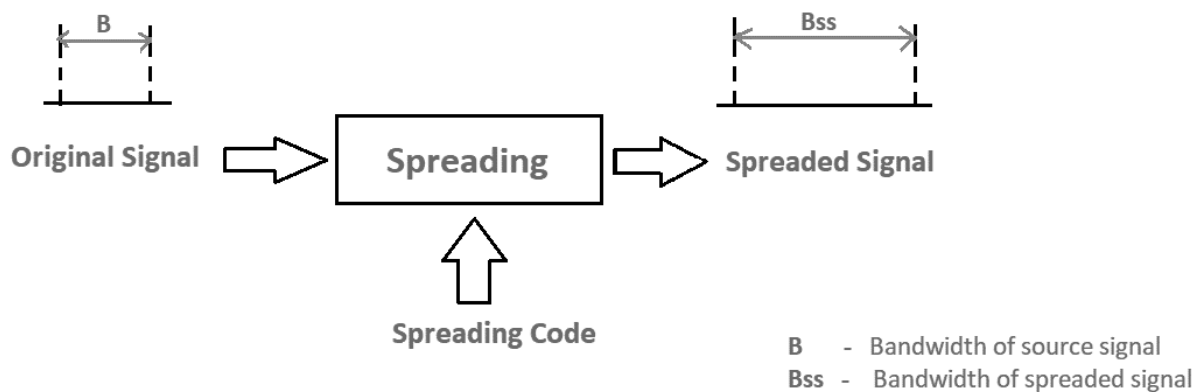
Telephone Company Site: DSLAM



The view is different from the telephone company site. Instead of using an ADSL modem, they are using a DSLAM, i.e., Digital Subscriber Line Multiplex. In addition to ADSL features, it has an additional feature to packetize the data to be sent to the internet.

Spread Spectrum

Spread Spectrum is a wireless communication method where the signal is spread over a wider frequency range than needed. This helps make the communication more secure and reliable by protecting it from noise, hacking, and signal blocking. The 'spread code' is a patterned series of numbers that enlarges the original signal's bandwidth. This method is typically used in situations where secure transmission is crucial.



- The spread spectrum is a type of modulation where the modulated signal bandwidth is much larger than the baseband signal bandwidth, i.e., spread spectrum is a wideband scheme.
- A special code (pseudo noise) is used to spread the signal, and the same code is needed at the receiver to get the original signal back.

1. Direct Sequence Spread Spectrum (DSSS)

DSSS is a transmission technology used in wireless communication. This method involves spreading the signal over a larger bandwidth than what is required for simple transmission. The original data signal is combined with a higher-rate bit sequence, also known as a chipping code or pseudo noise (PN) code, to increase the signal's bandwidth.

At the receiving end, the signal is demodulated with the PN code, and the original message is restored.

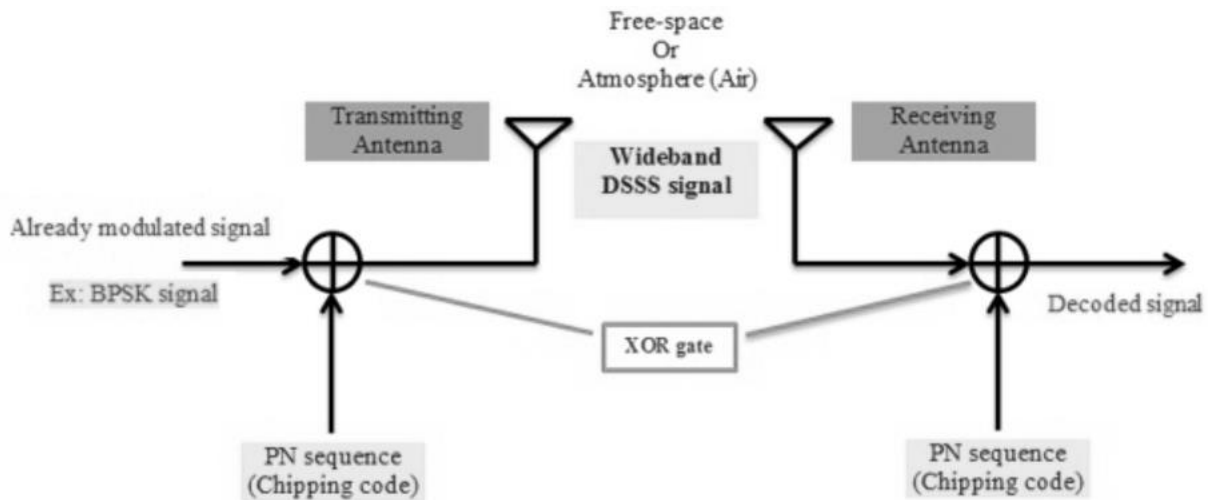


Figure: DSSS Communication System

- DSSS is the most widely used technology for spread spectrum.
- PN (Pseudo-noise) sequence is used to modulate the already modulated signal. Pseudo means semi-random.
- Transmitted signal bandwidth is greater than information signal bandwidth.
- XOR technique is used to combine input data stream with PN sequence
- PN codes produced by the transmitter are already known by the receiver.

Input Bit stream A	0	1	0	0	1	0	1	1
PN sequence B	0	1	1	0	1	0	0	1
Transmitted signal C	0	1	1	0	0	1	1	0
Received signal C	0	1	1	0	0	1	1	0
PN sequence B	0	1	1	0	1	0	0	1
Output Bit stream A	0	0	0	0	1	1	1	1

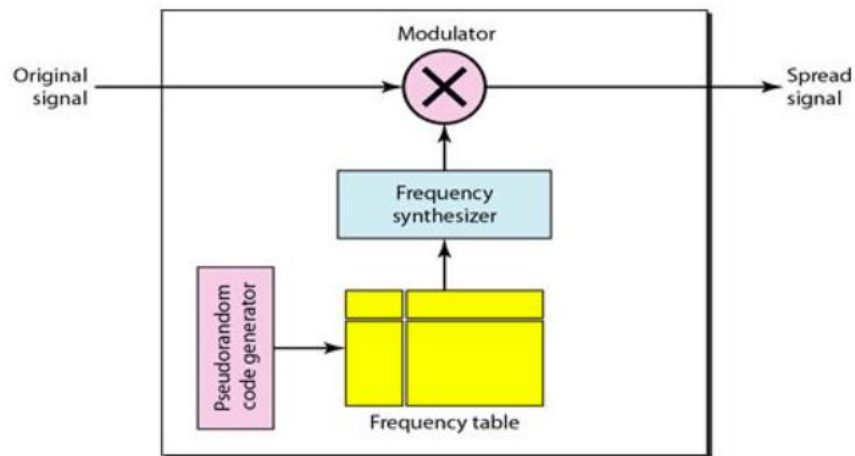
Decoded output → same as Input Bit stream

2. Frequency Hopping Spread Spectrum (FHSS)

Frequency Hopping Spread Spectrum (FHSS) is a method of transmitting radio signals by rapidly switching a carrier among many frequency channels, using a pseudorandom sequence known to both the transmitter and the receiver.

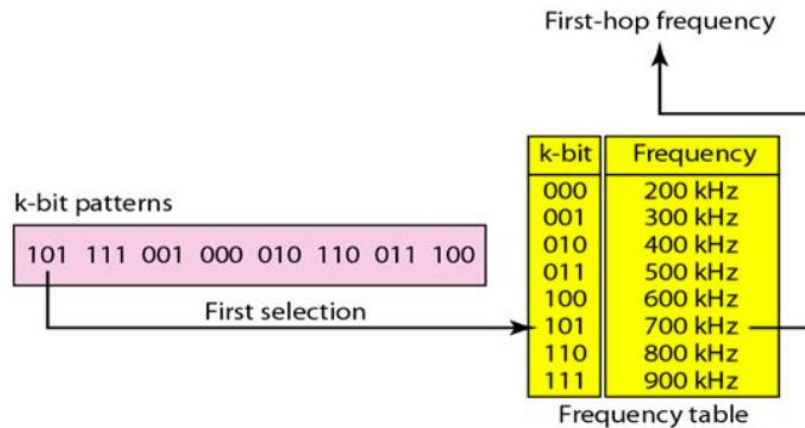
The data signal is modulated with a narrowband carrier signal that “hops” in a random but predictable sequence from frequency to frequency as a function of time over a wide band of frequencies. It is used as a multiple access method in the frequency-hopping code division multiple access (FH-CDMA) scheme.

The following figure shows the general layout for FHSS. A pseudorandom code generator, called pseudorandom noise (PN), creates a k-bit pattern for every hopping period.



The frequency table uses the pattern to find the frequency to be used for this hopping period and passes it to the frequency synthesizer. The frequency synthesizer creates a carrier signal of that frequency, and the source signal modulates the carrier signal.

For Example, M is 8 and k is 3. The pseudorandom code generator will create eight different 3-bit patterns. These are mapped to eight different frequencies in the frequency table as shown in the following figure.



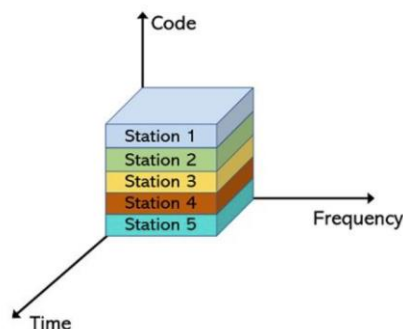
The pattern for this station is 101, 111, 001, 000, 010, and 100. Note that the pattern is pseudorandom; it repeats after eight hops. This means that at hopping period 1, the pattern is 101. The frequency selected is 700 kHz; the source signal modulates this carrier frequency.

The second k-bit pattern selected is 111, which selects the 900-kHz carrier; the eighth pattern is 100, the frequency is 600 kHz. After eight hops, the pattern repeats, starting from 101 again.

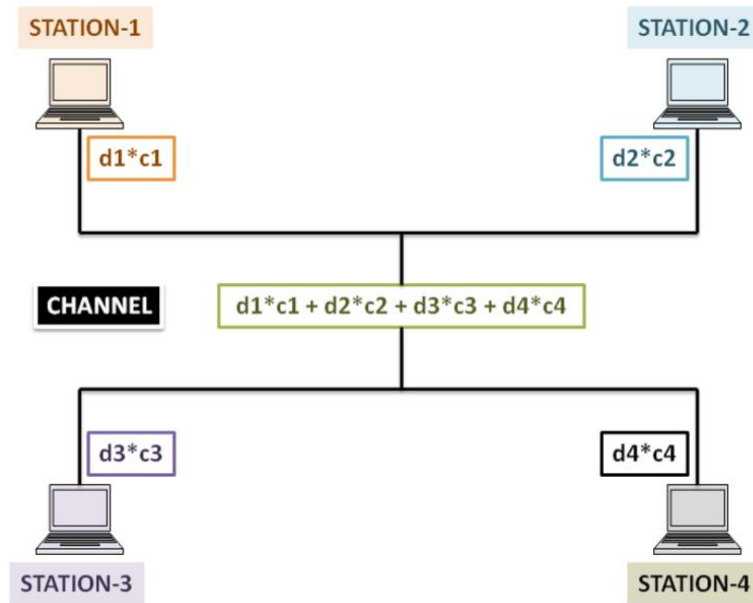
3. Code Division Multiple Access (CDMA)

CDMA is a technique that allows multiple users to simultaneously transmit data signals over a common channel by assigning unique spreading codes to each individual user. This leads to the increasing of bandwidth used by transmitting stations from a few Hz to a few MHz.

CDMA uses the principle of spread spectrum. Due to this, the various signals are modulated after which a single signal is transmitted and is correlated at the receiving end using the spreading function to get the actual data.



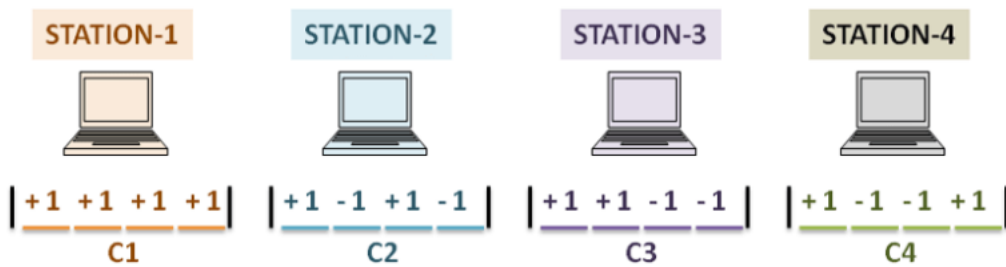
The diagram below explains the CDMA method.



- There are four stations 1, 2, 3, 4, as shown in figure above. Station-1, Station-2, Station-3, and Station-4 have $d1$, $d2$, $d3$, and $d4$ data, and codes $c1$, $c2$, $c3$ and $c4$, respectively.
- Here, we have assumed that if we multiply each code by the other, we get zero, and if we multiply each code itself, we get 4, which is the number of stations.
- Now, station-1 multiplies its data $d1$ by the code $c1$, which makes $d1*c1$. Similarly, station-2, station-3 and station-4 form $d2*c2$, $d3*c3$, and $d4*c4$ respectively.
- So, the sum of all the stations' data will be broadcast on a single channel.
- Suppose station-1 and station-2 are communicating, and station-2 wants to hear what station-1 is saying. It multiplies the data by the code $c1$ of the station-1 on the channel. Then station-2 will divide the result by 4 to get the data from station-1.

As we learned that CDMA is a coding-based principle in which a code is assigned to each station. Codes are a sequence of numbers, known as chips.

The diagram below explains the chip sequence:



As shown in figure, each code consists of a sequence of numbers know as chips. In addition, the chips are not chosen at random. Chip sequence are called orthogonal sequences and are generated by the [Walsh table](#).

The chips have the following properties:

- In each code, the sequence is made up of N (Number of stations) elements
- If we multiply two identical sequence elements, we get the total number of stations present on the channel.

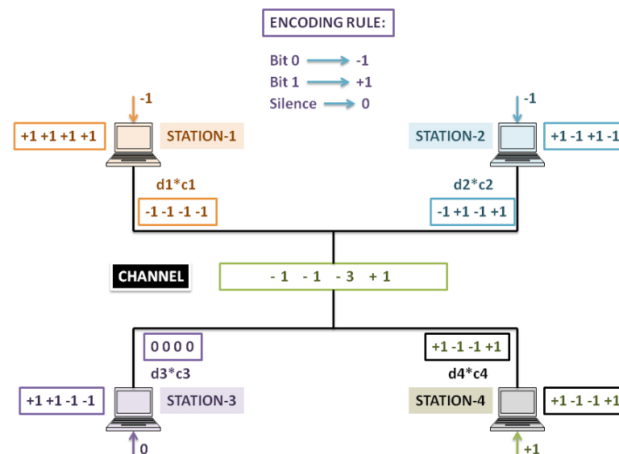
Example: $[+1 \ +1 \ +1 \ +1] * [+1 \ +1 \ +1 \ +1] = [+4 \ +4 \ +4 \ +4]$

- If we multiply and sum two different chip sequence element by element, we get 0.

Example: $[+1 \ +1 \ +1 \ +1] * [+1 \ -1 \ +1 \ -1] = 1-1+1-1 = 0$

CDMA encoding and decoding

The encoding method describes the rules for encoding data. If the station sends a bit 0 and 1, the method encodes it as -1 and +1 respectively. When a channel is idle, it does not send any signal displayed as 0.



- As shown in figure above, there are four stations, station-1, station-2, station-3 and station-4, communicating on a common channel.
- Suppose station-1 and station-2 are sending 0 bits and station-2 is sending 1 bit, but station-3 is silent. Therefore, the data 0, 0, silent, 1 will be encoded as -1, -1, 0, +1 respectively, according to the encoding rule.
- After that, each station multiplies the number corresponding to its chip, which is unique to a station. The sequence [-1 -1 -3 +1] generated on a channel is the sum of all four sequences.
- Now suppose station-3 is listening to station-2 and wants to know what station-2 is saying, then it will decode the data. So it multiplies the total data on the channel by the code of station-2 and divides the result by the number of stations (N).

$$[-1 -1 -3 +1] * [+1 -1 +1 -1] = -1+1-3-1 = -4$$

$$-4/4 = -1$$

Walsh Table

In CDMA, the chip sequence are chosen pseudo randomly. The sequences are called orthogonal sequences. Walsh table is a two dimensional table with an equal number of rows and columns used to generate a kind of chip sequence for CDMA, where each row is a sequence of chips.

The Walsh table W_1 for a one-chip sequence has one row and one column. We can choose -1 or +1 for the chip. According to Walsh, if we know the table for N sequences W_N , we can create the table for $2N$ sequences W_{2N} .

The W_N with the overbar stands for the complement of W_N where each +1 is changed to -1 and vice versa.

$$W_1 = \begin{bmatrix} +1 \end{bmatrix}$$

$$W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

Let us see how we can create W_2 and W_4 from W_1 .

- After we select W_1 , W_2 can be made from four W_1 's with the last one the complement of W_1 .
- After W_2 is generated, W_4 can be made of four W_2 's, with the last one the complement of W_1 .

$$W_1 = \begin{bmatrix} +1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix}$$

$$W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

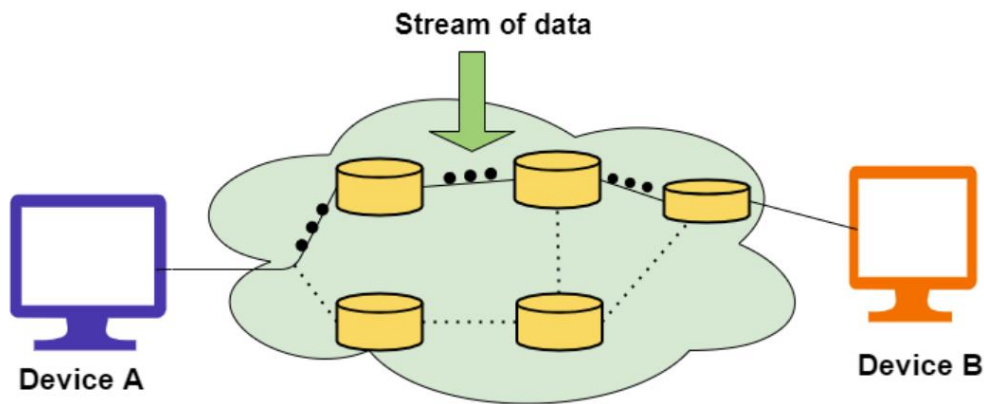
Switching in Networking

Switching is a process of transferring data packets from one device to another in a network using specific devices called switches. Data packets are transferred utilizing switches from one network to another or from one device to another during the process of switching. A switch is a piece of hardware that operates at the OSI model's data link layer, i.e., layer 2. Incoming data packets from a source device or network are largely handled by a switch, which also selects the best port through which the data packets will go to their destination device or network.

Switching in networking is mainly divided into two parts:

A. Connection-oriented

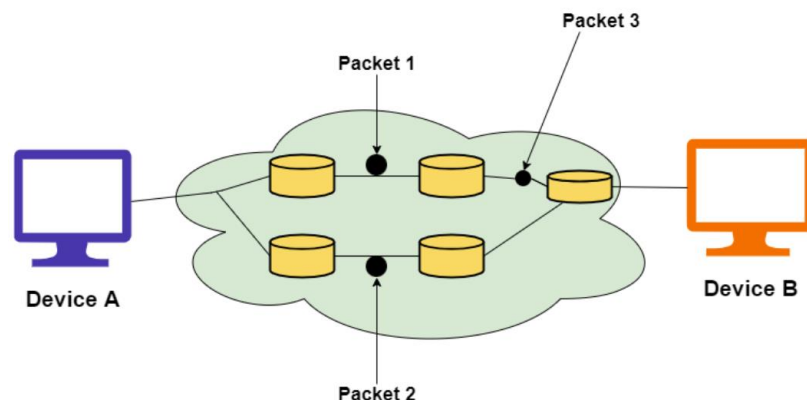
Connection-oriented switching first requires a dedicated path or a prior agreement between the source and the destination. Once the dedicated path is set, data is sent in the form of a continuous stream of bits or bytes. Afterwards, Data is transmitted along the same path by the switch.



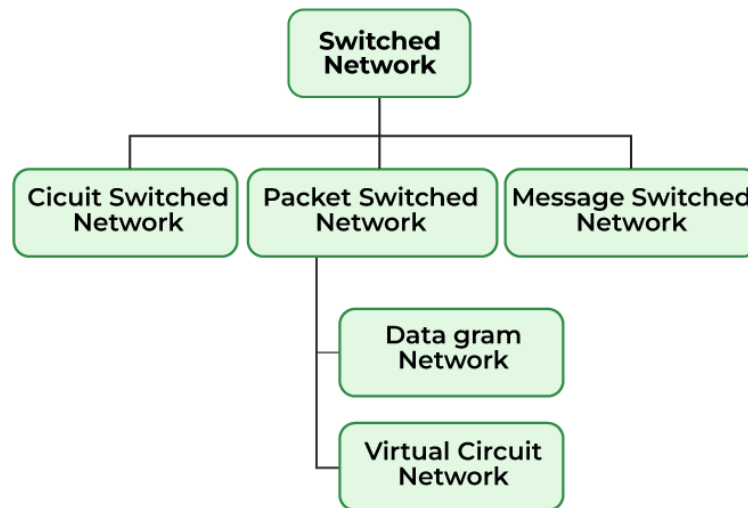
In the above diagram, a solid line between Device A and Device B indicates that there is a dedicated link with which the stream of data travels between them. With the help of this link, a receiver can send an acknowledgment to the sender about the status of the packet.

B. Connectionless

With connectionless switching, there is no need for a dedicated path or any prior agreement between the source and the destination. In this, the data is divided into smaller components known as packets, which are transmitted individually on the basis of the destination address.

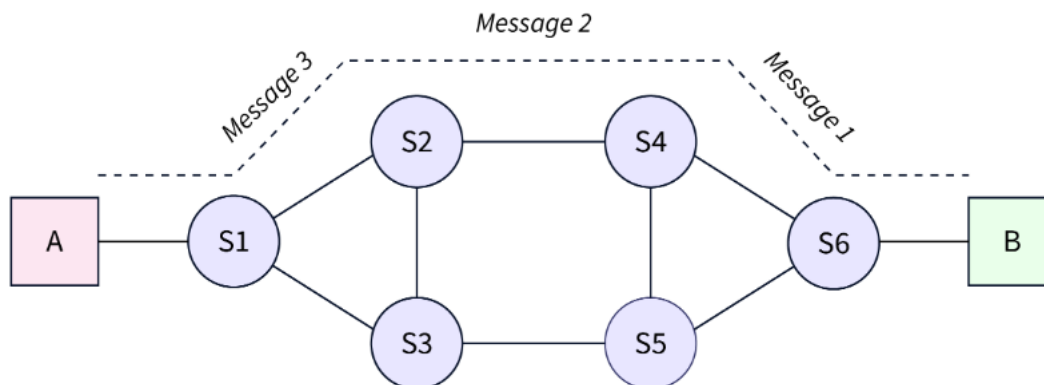


In connectionless the data is transferred in one direction from source to destination without checking the destination is still there or not or if it prepared to accept the message. Authentication is not needed in this. An example of a Connectionless service is **UDP (User Datagram Protocol) protocol**.



1. Circuit Switching

Circuit Switching is a type of switching in which a connection is established between the source and destination before communication. This connection receives the complete bandwidth of the network until the data is transferred completely. However, circuit switching can be inefficient and costly due to its requirement for dedicated resources, making it less suitable for high-traffic or large-scale networks.



When a user wants to send data, audio, or video, a request signal is sent to the receiver, and the receiver responds with an acknowledgment to ensure that the dedicated path is available. After receiving acknowledgment, the data is sent along a predefined path. Previously circuit

switching was used in public telephone networks for voice transmission, but in the current scenario, it is replaced by virtual circuits.

It involves three phases:

- **Circuit Establishment**
 - This phase is also called as setup phase.
 - In this phase, a dedicated connection is established directly between two nodes or from the source to the destination using a series of switching centers.
 - The sender at the source and the receiver at the destination send communication signals to request and acknowledge the establishment of the circuit.
- **Data Transfer**
 - Data can be transferred between the source and destination once the circuit has been established. The link between the two parties remains as long as they communicate.

- **Connection Disconnect**

Disconnection in the circuit occurs when one of the users initiates the disconnect. When the disconnection occurs, all intermediary linkages between the sender and receiver are terminated.

Advantages of Circuit Switching

- **Guaranteed Data Rate:** A fixed channel ensures consistent speed.
- **No Delay:** A Dedicated path allows continuous data flow.
- **Reliable:** A Stable connection reduces data loss or errors.
- **Quality of Service:** Prioritizes traffic like voice and video.
- **Secure:** Only the two users can access the channel.
- **Easy Management:** Pre-set path simplifies control.
- **Compatible:** Works with many devices and protocols.

Disadvantages of Circuit Switching

- **Limited Scalability:** Hard to manage in large networks.
- **Vulnerable to Failures:** A path failure breaks communication.
- **Low Flexibility:** The Circuit can't be reused during communication.
- **Resource Wastage:** Reserves bandwidth even when idle.
- **Expensive Setup:** Costly to install and maintain.
- **Failure Prone:** The Entire connection fails if path breaks.
- **Not for High Traffic:** Inefficient for bursty data flows.
- **Delay in Setup:** Time needed to establish a path adds latency.

2. Packet Switching

In packet switching, the entire message to be sent is divided into multiple smaller packets. This process of dividing a single message into smaller packets is called **packetization**. These smaller packets are sent after the other. It gives the advantage of pipelining and reduces the total time taken to transmit the message.

[For more detail please refer to the following pdf:](#)



Packet Switching.pdf

3. Message Switching

[Please refer to the following pdf:](#)



Message
Switching.pdf